Introduction to ∞-operads

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Lecture 1. 1. From operads to ∞-operads

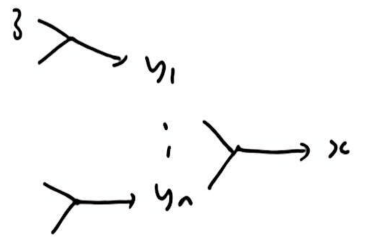
Operads

Convention: operad = symmetric many-object / coloured operad in sets

Idea. An operad is a structure like a category but mors are allowed to have several inputs, and can permute these.

Def. An operad $\mathcal{O}$ has

- set of objects / colours $\mathrm{ob}\,\mathcal{O}$
- for $(\overbrace{y_1, \dots, y_n}^{y})$, x, $y_i \in \mathrm{ob}\,\mathcal{O}$, a set $\mathcal{O}((y_1, \dots, y_n), x)$ of multimorphisms.
- composition $f : y \to x$
  $z_1 \xrightarrow{g_1} y_1, \dots, z_n \to y_n$



$G = (g_1, \dots, g_n)$, $f \circ G : z \to x$

- $\mathrm{id}_x : (x) \to x$
- $\sigma \in \Sigma_n$, $\mathcal{O}((y_1, \dots, y_n), x) \xrightarrow{\sim} \mathcal{O}((y_{\sigma(1)}, \dots, y_{\sigma(n)}), x)$
  compatible w/ mult. in Σ_n

s.t. composition is associative $(f \circ G) \circ (H_1, \dots, H_n) = f \circ (g_1 \circ H_1, \dots, g_n \circ H_n)$
unital $\mathrm{id}_x \circ f = f$, $f \circ (\mathrm{id}_{y_1}, \dots, \mathrm{id}_{y_n}) = f$, compatible w/ Σ_n-actions

Notation If $\mathrm{ob}\,\mathcal{O} = \{*\}$, write $\mathcal{O}(n) = \mathcal{O}((\underbrace{*, \cdots *}_{n}), *)$,

$\mathcal{O}(n)$ has Σ_n-action, comp. looks like

$$\mathcal{O}(k) \times \mathcal{O}(n_1) \times \cdots \times \mathcal{O}(n_k) \longrightarrow \mathcal{O}(n_1 + \cdots + n_k)$$

Examples

- Comm has a single obj, $\mathrm{Comm}(n) = *$, $\forall n$
- Assoc — " —, $\mathrm{Assoc}(n)$ = set of orderings of $1, \cdots, n$ w/ free Σ_n-actions

  Comp. is concatenating orderings

- CM has two objects a, m

$$CM(Y, x) = \begin{cases} *, & Y = (a, \cdots a), \quad x = a \\ *, & Y = (a, \cdots, a, m) \text{ up to permutation}, \quad x = m \\ \phi, & \text{otherwise} \end{cases}$$

- BM has objs l, m, r.

$$BM(X, l) = \{\text{isoms } X \simeq (l, \cdots, l)\}$$
$$BM(X, r) = \{\text{isoms } X \simeq (r, \cdots, r)\}$$
$$BM(X, m) = \{\text{isoms } X \simeq (l, \cdots, l, m, r, \cdots, r)\}$$

Example $(\mathcal{V}, \otimes)$ a sym. mon. cat, $\mathcal{V}_{\mathrm{opr}}$ has same obj. as $\mathcal{V}$, w/

$$\mathcal{V}_{\mathrm{opr}}((x_1, \cdots x_n), y) = \mathcal{V}(x_1 \otimes \cdots \otimes x_n, y)$$

Def $\mathcal{O}, \mathcal{P}$ operads. a **functor** $\mathcal{O} \xrightarrow{F} \mathcal{P}$ assigns $F(x) \in \mathcal{P}$ for $x \in \mathcal{O}$.

$\mathcal{O}((x_1, \cdots, x_n), y) \longrightarrow \mathcal{P}((Fx_1, \cdots Fx_n), Fy)$ compatible w/ composition, id's, Σ_n-actions.

Prop

Opd = cat of operads & functors.

Functors $\mathcal{O} \to \mathcal{V}_{opr}$ are <u>$\mathcal{O}$-algebras</u> in $\mathcal{V}$

<u>Eg</u> – A Comm-alg in $\mathcal{V}$ is a comm alg. obj. in $\mathcal{V}$

$$\begin{aligned} \mathrm{Comm} &\longrightarrow \mathcal{V}_{opr} \\ * &\longmapsto A \\ \cdot \in \mathrm{Comm}(0) &\mapsto \mathbb{1} \to A \\ \cdot \in \mathrm{Comm}(2) &\mapsto A \otimes A \to A \end{aligned}$$

– Assoc-alg in $\mathcal{V}$ = associative alg

– $* \longmapsto A \in \mathcal{V}$

– $\mathrm{Assoc}(0) \longmapsto \mathbb{1} \to A$

$\mathrm{Assoc}(2)$

$$\begin{aligned} 1<2 \quad & A \otimes A \longrightarrow A \\ 2<1 \quad & A \otimes A \longrightarrow A \end{aligned}$$

– CM-alg in $\mathcal{V}$ = comm. alg w/ module

– BM-alg is two assoc. algs & bimodules

$\mathcal{W}$ sym. mon., $\mathcal{W}_{opr}$-alg in $\mathcal{V}$ is a lax sym. mon. functor $\mathcal{W} \xrightarrow{F} \mathcal{V}$.

multimorphism in $\mathcal{W}_{opr}$ factors

$$\begin{tikzcd} (w_1, \dots, w_n) \arrow[r] \arrow[dr] \arrow[dd, mapsto] & w' \\ & w_1 \otimes \dots \otimes w_n \arrow[u] \\ (Fw_1, \dots, Fw_n) \arrow[r] \arrow[dr] & Fw' \\ & F(w_1 \otimes \dots \otimes w_n) \arrow[u] \end{tikzcd} \longrightarrow \begin{array}{c} Fw_1 \otimes \dots \otimes Fw_n \\ \downarrow \\ F(w_1 \otimes \dots \otimes w_n) \end{array}$$

Operads via categories of operators

Notation. $\mathbb{F}$ = cat. of finite sets. $\mathbb{F}_*$ = finite pointed sets

$\langle n\rangle = (\{0,1,\cdots,n\},0)$. $\mathbb{F}_*$ is also cat. of finite sets w/ partially defined functions

$$S \hookleftarrow S_0 \longrightarrow T \hookleftarrow T_0 \longrightarrow$$

$$\updownarrow$$

$$S_* \longrightarrow T_*$$

$$x \notin S_0 \longmapsto \text{base pt}$$

Def (May–Thomason) $\mathcal{O}$ an operad. Its <u>cat. of operators</u>

$Op(\mathcal{O}) \longrightarrow \mathbb{F}_*$ has

- obj. $(\langle n\rangle, x_1,\cdots x_n)$, $x_i \in$ ob $\mathcal{O}$
- mors $(x_1,\cdots,x_n) \longrightarrow (y_1,\cdots,y_m)$ are

  $\langle n\rangle \xrightarrow{\varphi} \langle m\rangle$ & for $i=1,\cdots,m$, a multimor

  $$(x_j)_{j\in\varphi^{-1}(i)} \longrightarrow y_i$$

Composition: $(x_1,\dots,x_n) \xrightarrow{(\varphi,\ \Phi_i)} (y_1,\cdots,y_m) \xrightarrow{(\psi,\ \Psi_j)} (z_1,\cdots,z_r)$

$\psi\varphi$,

$$(x_k)_{k\in(\psi\varphi)^{-1}(i)} \xrightarrow{(\Phi_i)} (y_i)_{i\in\psi^{-1}(j)} \xrightarrow{\Psi_j} z_j \quad \leftsquigarrow \text{composition in } \mathcal{O}$$

A functor of operads $\mathcal{O}\longrightarrow\mathcal{P}$ gives

$$\begin{tikzcd} Op(\mathcal{O}) \arrow[rr] \arrow[dr] & & Op(\mathcal{P}) \arrow[dl] \\ & \mathbb{F}_* & \end{tikzcd}$$

$\rightsquigarrow$ functor $Opd \xrightarrow[Op]{} Cat/\mathbb{F}_*$

Def $\varphi : \langle n \rangle \to \langle m \rangle$ in $\mathbb{F}_*$ is <u>active</u> if $\varphi^{-1}(0) = \{0\}$.

$$(\underline{n} = \underline{n} \to \underline{m})$$

<u>inert</u> if φ is an iso. away from base pt

$$|\varphi^{-1}(i)| = 1 \quad \text{for } i \neq 0 \qquad (\underline{n} \hookleftarrow \underline{n_0} = \underline{n_0})$$

This is a fact. System on $\mathbb{F}_*$

$$p_i : \langle n \rangle \longrightarrow \langle 1 \rangle \qquad j \longmapsto \begin{cases} 0, & j \neq i \\ 1, & j = i \end{cases} \qquad (\underline{n} \hookleftarrow \{1\} \to 1)$$

How does $Op(\mathcal{O})$ encode str. of $\mathcal{O}$?

- Unique active map $\langle n \rangle \xrightarrow{\alpha_n} \langle 1 \rangle$
  - mor in $Op(\mathcal{O})$ over α_n is an n-ary multimor in $\mathcal{O}$
  - For $\langle n \rangle \xrightarrow{\alpha} \langle m \rangle$, mor over α is a list of m multimors w/ input of size $n_i = \alpha^{-1}(i)$.
  - Comp. over $\langle n \rangle \xrightarrow{\alpha} \langle m \rangle \xrightarrow{\alpha_m} \langle 1 \rangle$ gives multimor comp. in $\mathcal{O}$

  Inert maps allow us to recover these list decompositions just from the category $Op(\mathcal{O}) \longrightarrow \mathbb{F}_*$

Def. $p : E \to B$. A mor $f : x \to y$ in E is <u>p-cocartesian</u> if $\forall z \in E$, the square

$$\begin{tikzcd} E(y,z) \arrow[r, "f^*"] \arrow[d] & E(x,z) \arrow[d] \\ B(py,pz) \arrow[r, "p(f)^*"] & B(px,pz) \end{tikzcd}$$

is a pull back.

$$\begin{tikzcd} & y \arrow[dr, dashed, "\exists"] & \\ x \arrow[ur, "f"] \arrow[rr] & & z \end{tikzcd}$$

$$\begin{tikzcd} & py \arrow[dr] & \\ px \arrow[ur, "pf"] \arrow[rr] & & pz \end{tikzcd}$$

E has p-cocartesian lifts of a set S of mor on B if $\forall x \in E$,

$px \xrightarrow{f} b$ in S, $\exists$ a p-cocart mor $\bar{f} : x \to y$, $p(\bar{f}) = f$.

p is a <u>cocartesian fibration</u> (+ Grothendieck opfibration) if E has p-cocart. lifts of all mors. p is an <u>isofibration</u> if it has them for isos

<u>Prop</u> An isofib $p : E \to \mathbb{F}_*$ is equiv. (over $\mathbb{F}_*$) to cat of op's of an operd $\mathcal{H}$

(1) E has p-cocart. lifts of inert mors

(2) Given $X \in E_{\langle n \rangle}$, cocart mors $X \to X_i$ over ρ_i, $Y \in E_{\langle m \rangle}$

$$\begin{tikzcd} E(Y,X) \arrow[r] \arrow[d] & \prod_{i=1}^{n} E(Y,X_i) \arrow[d] \\ \mathrm{Hom}_{\mathbb{F}_*}(\langle m \rangle, \langle n \rangle) \arrow[r] & \prod_{i=1}^{n} \mathrm{Hom}_{\mathbb{F}_*}(\langle m \rangle, \langle 1 \rangle) \end{tikzcd}$$

is a pullback

(3) Given $X_1, \dots, X_n$ in $E_{\langle 1 \rangle}$, $\exists X \in E_{\langle n \rangle}$ w/ cocart. mors $X \to X_i$ over ρ_i

A functor
$$\begin{tikzcd} E \arrow[rr, "F"] \arrow[dr] & & E' \arrow[dl] \\ & \mathbb{F}_* & \end{tikzcd}$$
comes from a functor of operads iff

F preserves inert cocart. mors.

<u>Upshot</u> Op is equiv. to subcat. of $\mathrm{Cat}/\mathbb{F}_*$ w/ objs isofib. w/ (1) - (3),

& morphisms preserving inert cocarts.

Lecture 2. ∞-operads as dendroidal Segal spaces

∞-cats as Segal spaces.

Can view Δ as full subcat of Cat on $[n] = 0 \to 1 \to \dots \to n$

$\rightsquigarrow \quad N : \mathrm{Cat} \longrightarrow \mathrm{Fun}(\Delta^{op}, \mathrm{Set})$

$NC_n = \mathrm{Hom}([n], C)$ = set of n composable mors in C.

this is fully faithful, can characterise image as X s.t.

$$X_n \xrightarrow{\sim} X_1 \times_{X_0} X_1 \times_{X_0} \dots \times_{X_0} X_1 \quad \text{via} \quad \{i\}, \{i-1, i\} \hookrightarrow [n]$$

X_0 = obs

X_1 = mors

$$\begin{tikzcd} X_1 \arrow[r, shift left=2, "d_0"] \arrow[r, shift right=2, "d_1"'] & X_0 \arrow[l, "id"' description] \end{tikzcd}$$

$$X_1 \times_{X_0} X_1 \xleftarrow{\sim} X_2 \xrightarrow{d_1} X_1$$

compose

Why Δ? Obs are "shapes we can compose in a cat"

morphisms encode both

- composition operation $[1] \to [n]$
  $0 \mapsto 0$
  $1 \mapsto n$
- decomposition of shapes $[1] \to [n]$
  $0 \mapsto i-1$
  $1 \mapsto i$

Def (Rezk) A **Segal space** is a functor $X : \Delta^{op} \longrightarrow \mathcal{S}$ — ∞-cat of spaces/∞-gpds

s.t. $X([n]) \xrightarrow{\sim} X([1]) \times_{X([0])} X([1]) \times_{X([0])} \dots \times_{X([0])} X([1])$

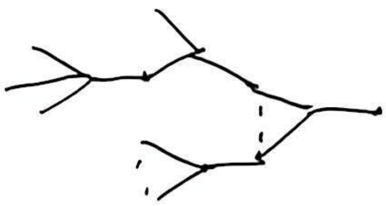
Can define fully faithful & ess. surj. mors of Segal spaces, want to invert to get Cat_∞.

Rezk showed this loc'n by restricting to full subcat of <u>complete</u> Segal spaces.

$\lnot$

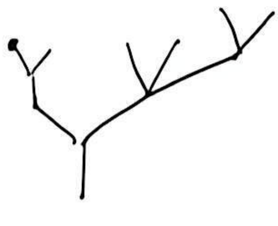
The dendroidal cat

Basic composable shapes for operads are <u>trees</u>.



<u>Idea</u>

1. define the relevant type of trees + subtree inclusion
2. define a "free operad" on tree
3. define Ω as full subcat of Opd on these
4. describe maps
5. describe (∞-) operads as presheaves on Ω.



A tree is described by

$$E \xleftarrow{i} V_1 \xrightarrow{u} V \xrightarrow{o} E \qquad (*)$$

E = set of edges

V = set of vertices

V_* = set of vertices + incoming edge

$i = (v, e) \longmapsto e$

$u = (v, e) \longmapsto v$

o = v maps to its unique outgoing edge

<u>Def</u> (Rezk) A <u>tree</u> is a diagram $(*)$ of finite sets s.t.

- o is injective

- i' is injective & $\exists !$ edge R not in image of i

- define $\sigma : E \to E$ by $\sigma(R) = R$, $\sigma(x) = o\,u(i^{-1}x)$

  $\forall\, x \in E$, we have $\sigma^k x = R$ for some k.

A <u>subtree inclusion</u> is a diagram

$$\begin{tikzcd} E \arrow[d, "f"'] & V_* \arrow[l] \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & V \arrow[r] \arrow[d, "g"] & E \arrow[d, "f"] \\ E' & V'_* \arrow[l] \arrow[r] & V' \arrow[r] & E' \end{tikzcd}$$

pullback

f and g are automatically injective.

Given a tree T, define an operad $\Omega[T]$ w/ objects = edges of T

& multimors $(x_1, \dots, x_n) \mapsto y$ = subtrees of T w/ $(x_1, \dots, x_n)$ as leaves & y as root

- composition = gluing trees together.

Ω = full subcat of Opd on these objects.

$T \rightsquigarrow sub(T)$ = set of subtrees in T

$sub_*(T)$ = set of subtrees in T w/ a marked leaf

$$E \leftarrow sub_*(T) \to sub(T) \xrightarrow{root} E$$

Can describe maps in Ω as

T w/ edges E

T' w/ edges E'

$$\begin{tikzcd} E \arrow[d] & sub_*(T) \arrow[l] \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & sub(T) \arrow[r] \arrow[d] & E \arrow[d] \\ E' & sub_*(T') \arrow[l] \arrow[r] & sub(T') \arrow[r] & E' \end{tikzcd}$$

- this is determined by where we send vertices in T

T_0 subtree of T has to go to the subtree of T' obtained by gluing the subtrees assigned to vertices in T_0.

$$\begin{tikzcd} E \arrow[d] & V_* \arrow[l] \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & V \arrow[r] \arrow[d] & E \arrow[d] \\ E' & \mathrm{Sub}_*(T') \arrow[l] \arrow[r] & \mathrm{Sub}(T') \arrow[r] & E' \end{tikzcd}$$

A mor. in Ω is <u>inert</u> if it is a subtree incl.

& <u>active</u> if it preserves the boundary $\left(\begin{array}{ccc} \text{leaves} & \mapsto & \text{leaves} \\ \text{root} & \mapsto & \text{root} \end{array}\right)$

A mor. in Ω factors uniquely up to isom. as an active map followed by an inert map

<u>Dendroidal Segal spaces</u>

<u>Def</u> An <u>algebraic pattern</u> is an ∞-cat. $\mathcal{O}$ w/ fact system $(\mathcal{O}_{int}, \mathcal{O}_{act})$ s.t. any mor factors as $\xrightarrow{\text{inert}} \xrightarrow{\text{active}}$ + a collection of elementary objects $\mathcal{O}^{el} \subset \mathcal{O}_{int}$ (e.g. $\mathcal{O} = \Omega^{op}$, $\mathbb{F}_*$)

A <u>Segal object</u> for $\mathcal{O}$ in an ∞-cat is $\mathcal{O} \xrightarrow{F} \mathcal{C}$ s.t.

$\forall X \in \mathcal{O}$, $\left[\mathcal{O}^{el}_{X/} = \mathcal{O}^{el} \times_{\mathcal{O}^{int}} \mathcal{O}^{int}_{X/}\right.$, i.e. inert maps $X \to$ elt'ary$\left.\right]$

$(\mathcal{O}^{el}_{X/})^{\triangleleft} \to \mathcal{O} \xrightarrow{F} \mathcal{C}$ is a limit, i.e. $F(X) \simeq \lim\limits_{E \in \mathcal{O}^{el}_{X/}} F(E)$

<u>Example</u>. Δ has an (active, inert) fact system.

$\varphi : [n] \to [m]$ is <u>inert</u> if $\varphi(i) = \varphi(0) + i$

is <u>active</u> if $\varphi(0) = 0$, $\varphi(n) = m$

Δ^{op} is an algebraic pattern w/ this fact. system & $[0], [1]$ as elt'ry objects

$\rightsquigarrow$ Segal Δ^{op}-ob in $\mathcal{S}$ is Segal space

$F([n])$: lim over inert maps

$$[1] \to [n] \quad \longleftrightarrow \quad \{i-1, i\} \hookrightarrow [n]$$
$$[0] \to [n] \qquad\qquad \{i\} \to [n]$$

<u>Def</u>. Ω^{op} is an algebraic pattern w/ inert-active fact-system above,

elementary objects $\eta = | = * \leftarrow \phi \to \phi \to *$

$C_n =$  $= \underline{n+1} \hookleftarrow \underline{n} \to * \hookrightarrow \underline{n+1}$

A <u>dendroid Segal space</u> is a Segal Ω^{op}-ob. in $\mathcal{S}$, i.e.

$$F : \Omega^{op} \to \mathcal{S} \quad \text{s.t.} \quad F(T) \simeq \lim_{(E \rightarrowtail T)^{op}} F(E)$$

<u>eg.</u>

$F\Big($  $\Big) = F(C_0) \underset{F(\eta)}{\times} F(C_2) \underset{F(\eta)}{\times} F(C_3)$

$F(\eta) =$ sp. of objects

$F(C_n) =$ space of n-ary multimors

$F(T)$ = collection of multimors that can be composed in the shape of T

For any T w/ n-leaves, $\exists!$ active map $C_n \to T$

$\rightsquigarrow$ composition $F(T) \to F(C_n)$

Again can define FF, ES mors & invert those by restricting to full subcat of <u>complete</u> objects. (Cisinski–Moerdijk)

<u>Th</u>. $\mathrm{Opd}_\infty \simeq \mathrm{CSeg}_{\Omega^{op}}(\mathcal{S})$ (Hinich–Moerdijk)

$\Omega \in \mathrm{Opd}_\infty \rightsquigarrow \mathrm{Opd}_\infty \to \mathrm{PSh}(\Omega)$ – check this is fully faithful w/ image $\mathrm{CSeg}_{\Omega^{op}}(\mathcal{S})$

(1st comparison via Barwick's model of Segal pshs. Barwick, Chu–H.–Heuts)

Lecture 3. Back to categories of operators

Prop A functor $p: E \to \mathbb{F}_*$ is equiv. (over $\mathbb{F}_*$) to $Op(\mathcal{O})$ for $\mathcal{O} \in Opd$

iff (1) E has p-cocart lifts of inerts

(2) Given $X \in E_{\langle n \rangle}$, $X \to X_i$ over $\rho_i: \langle n \rangle \to \langle 1 \rangle$, $Y \in E_{\langle m \rangle}$,

$$\begin{tikzcd} E(Y, X) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \prod_{i=1}^{n} E(Y, X_i) \arrow[d] \\ \mathbb{F}_*(\langle m \rangle, \langle n \rangle) \arrow[r] & \prod_{i=1}^{n} \mathbb{F}_*(\langle m \rangle, \langle 1 \rangle) \end{tikzcd}$$

(3) Given $X_1, \dots, X_n \in E_{\langle 1 \rangle}$, $\exists X \in E_{\langle n \rangle}$ w/ $X \to X_i$ cocart. over ρ_i.

Note - cocart lifts of ρ_i give functors $E_{\langle n \rangle} \to \prod_{i=1}^{n} E_{\langle 1 \rangle}$

(3) $\Rightarrow$ ess. surj. (2) $\Rightarrow$ fully faithful, so this is an equiv.

- For $\varphi: \langle n \rangle \to \langle m \rangle$, $\rho_i \varphi: \langle n \rangle \to \langle 1 \rangle$

$$\begin{tikzcd} \langle n \rangle \arrow[dr, "\pi_i"'] \arrow[rr] & & \langle 1 \rangle \\ & \langle n_i \rangle \arrow[ur, "\alpha_{n_i}"'] & \end{tikzcd}$$

Cond (2) $\Leftrightarrow$ $E(Y, X)_\varphi \simeq \prod_{i=1}^{n} E(Y, X_i)_{\rho_i \varphi} \xleftarrow{\sim} \prod_{i=1}^{n} E(Y_i, X_i)_{\alpha_{n_i}}$

Have $Y \to Y_i$ over π_i

Idea of proof. Given $\mathcal{O}$, check $Op(\mathcal{O})$ satisfies (1)–(3).

(1) $\varphi: \langle n \rangle \to \langle m \rangle$ inert, $X = (x_1, \dots, x_n)$, $X \to X' = (x_{\varphi^{-1}(1)}, \dots, x_{\varphi^{-1}(m)})$

w/ multimors = identities is cocart. (Exer)

Conversely, given $p: E \to \mathbb{F}_*$, want to define operad $\mathcal{O}$

Ob $\mathcal{O}$ = ob $E_{\langle 1\rangle}$

A multimor $(x_1, \cdots x_n) \mapsto y$ is given by

$$\begin{array}{l} X \longrightarrow y \\ \cap \\ E_{\langle n\rangle} \end{array}$$ mor. in $\mathcal{O}$ over $\alpha_n: \langle n\rangle \longrightarrow \langle 1\rangle$

w/ $X \to x_i$ cocart over ρ_i (iso classes bc X not literally unique)

Given $X_1 \to y_1, \ \cdots, \ X_n \to y_n, \quad Y \to z$

$\langle m_1\rangle \to \langle 1\rangle \ \cdots \ \langle m_n\rangle \to \langle 1\rangle, \quad \langle n\rangle \to \langle 1\rangle$

find $\langle m\rangle = \langle m_1 + \cdots + m_n\rangle \xrightarrow{\pi_i} \langle m_i\rangle$

Find $X \to X_i$ cocart over π_i, build $X \to Y$ that decomposes to $X_i \to y_i$

compose $X \to Y \to z$.

<u>Upshot</u> Can identify Opd w/ a subcat of Cat$/\mathbb{F}_*$ w/ obs satisfying (1) – (3)

& mors preserve inert cocart mors.

<u>Prop.</u> For $p: E \to \mathbb{F}_*$, TFAE

(1) $E \simeq Op(\mathcal{V}_{opd})$ for $\mathcal{V}$ sym. mon.

$$\begin{tikzcd} E \arrow[dr] & & Op(\mathcal{V}_{opd}) \arrow[dl] \\ & \mathbb{F}_* & \end{tikzcd}$$

(2) p satisfies (1) – (3) & is a cocart fibration

(3) p is a cocart fib & $E_{\langle n\rangle} \xrightarrow{\sim} \hat{\prod_{i=1}} E_{\langle 1\rangle}$ via cocart lifts of ρ_i.

Moreover, strong sym. mon. functors correspond to functors over $\mathbb{F}_*$ that preserve

<u>ALL</u> cocart mors.

$\mathcal{O}p(\mathcal{O}) \longrightarrow \mathbb{F}_*$

$$\begin{tikzcd}[column sep=small] (x_1, \dots x_n) \arrow[rr] \arrow[dr] & & y \\ & \text{“}x_1 \otimes \cdots \otimes x_n\text{”} \arrow[ur] & \end{tikzcd}$$

mor. over $\alpha_n : \langle n \rangle \longrightarrow \langle 1 \rangle$

$$\text{“}x_1 \otimes \cdots \otimes x_n\text{”} = x_1 \otimes (x_2 \otimes \cdots)$$

$$\begin{tikzcd} (x, y) \arrow[r] \arrow[dr] & x \otimes y \arrow[d] \\ & z \end{tikzcd}$$

<u>Idea</u> Can regard this as giving a new def'n of operads

- concise & precise
- easy to use for many properties
- easy to generalize to ∞-cats.

$\mathcal{O}$ operad, when does it come from a sym. mon. cat?

- $\mathcal{O}((x_1, \dots x_n), y) \simeq \mathcal{O}(\otimes(x_1, \dots x_n), y)$

$$(x_1, \dots, x_n) \longrightarrow \otimes(x_1, \dots, x_n)$$

- $$\begin{tikzcd} (X_1, \dots, X_n) \arrow[r] \arrow[dr] & (\otimes X_1, \dots, \otimes X_n) \arrow[r] & \otimes(\otimes X_1, \dots, \otimes X_n) \\ & \otimes(X_1, \dots X_n) \arrow[ur, "\sim"] & \end{tikzcd}$$

<u>Lurie's ∞-operads</u>

<u>Def</u> An <u>∞-operad</u> is a functor of ∞-cats $p : \mathcal{O} \longrightarrow \mathbb{F}_*$ s.t.

(1) $\mathcal{O}$ has p-cocart lifts of inert mors

(2) The squares of mapping spaces as before are pullbacks

(3) Given $X_1, \dots, X_n \in \mathcal{O}_{\langle 1 \rangle}$, $\exists\, X \in \mathcal{O}_{\langle n \rangle}$ w/ $X \to x_i$ cocart over ρ^i.

A mor of ∞-operads is a comm. triangle
$$\begin{tikzcd} \mathcal{O} \arrow[rr, "F"] \arrow[dr] & & \mathcal{P} \arrow[dl] \\ & \mathbb{F}_* & \end{tikzcd}$$

s.t. F preserves inert cocart. mors

= $\mathcal{O}$- algebra in $\mathcal{P}$.

$\rightsquigarrow$ $Opd_\infty \subset Cat_{\infty / \mathbb{F}_*}$ subcat of ∞-opds & mors thereof.

$Alg_{\mathcal{O}}(\mathcal{P}) \subset Fun_{/\mathbb{F}_*}(\mathcal{O}, \mathcal{P})$ full subcat of $\mathcal{O}$-algs

<u>Example</u> $\mathcal{O}$ an operad in sets, then $Op(\mathcal{O}) \to \mathbb{F}_*$ is an ∞-opd

<u>Eg.</u> for $\mathcal{O}$ = Comm, Assoc, CM, BM

<u>Def</u> (Segal) $\mathcal{C}$ an ∞-cat w/ finite products. A <u>commutative monoid</u> in $\mathcal{C}$ is $M: \mathbb{F}_* \to \mathcal{C}$ s.t. $M(\langle n \rangle) \xrightarrow[(M(\rho_1), \dots, M(\rho_n))]{\sim} \prod_{i=1}^{n} M(\langle 1 \rangle)$, $\forall n$.

A <u>symmetric monoidal ∞-cat</u> is a commutative monoid in Cat_∞.

$\langle 0 \rangle \to \langle 1 \rangle$

$\rightsquigarrow * \to M(\langle 1 \rangle)$

$\langle 2 \rangle \to \langle 1 \rangle$

$\rightsquigarrow M(\langle 1 \rangle)^2 \to M(\langle 1 \rangle)$

etc.

$Fun(\mathbb{F}_*, Cat_\infty) \simeq$ Cocart fibs $/ \mathbb{F}_*$

<u>Prop</u> $p: \mathcal{E} \to \mathbb{F}_*$ a cocart fib. Then p is an ∞-operad iff p is the unstraightening of a sym mon. ∞-cat.

<u>Notation</u> A sym. mon. ∞-cat w/ underlying ∞-cat. $\mathcal{C}$ will often be denoted $\mathcal{C}^\otimes \to \mathbb{F}_*$

We'll use $\mathcal{O} \to \mathbb{F}_*$ for general ∞-opds.

A <u>lax</u> sym. mon. functor is
$$\begin{tikzcd} \mathcal{C}^{\otimes} \arrow[rr, "F"] \arrow[dr] & & \mathcal{D}^{\otimes} \arrow[dl] \\ & \mathbb{F}_* & \end{tikzcd}$$
s.t. F preserves inert cocarts

<u>strong</u> if it preserves all cocart mors.

Cartesian sym. mon. ∞-cats

$\mathcal{C}$ an ∞-cat w/ finite products

- should exist a (unique) sym mon ∞-cat $\mathcal{C}^{\times} \to \mathbb{F}_*$ w/ $\times$ as tensor
- $\mathcal{O}$-algs in $\mathcal{C}^{\times}$ should have a simpler description as certain functors $\mathcal{O} \to \mathcal{C}$.

$\mathcal{O} \xrightarrow{F} \mathcal{C}^{\times}$

$$\begin{array}{ccccc} (x_1, \ldots x_n) \in \mathcal{O}_{\langle 1\rangle}^{\times n} \simeq \mathcal{O}_{\langle n\rangle} & & F(x_1, \ldots, x_n) & & F(x_1, \ldots x_n) \in \mathcal{C} \to F(y) \\ \Big\downarrow \text{ over } \begin{array}{c}\langle n\rangle \\ \downarrow \alpha_n \\ \langle 1\rangle\end{array} & \longmapsto & \text{IS} & & \wr\Big\downarrow \\ y & & (Fx_1, \ldots Fx_n) & \text{v.s.} & \prod F(x_i) \\ & & \downarrow & & \\ \text{(multimor)} & & Fx_1 \times \cdots \times Fx_n & & \\ & & \Big\downarrow \quad \Big] \text{ in } \mathcal{C} & & \\ & & Fy & & \end{array}$$

Def An <u>$\mathcal{O}$-monoid</u> in $\mathcal{C}$ is $M: \mathcal{O} \to \mathcal{C}$ s.t. for $X \in \mathcal{O}_{\langle n\rangle}$, $X \to x_i$ cocart over ρ_i, we have $M(X) \xrightarrow{\sim} \prod M(x_i)$.

Def A sym. mon. ∞-cat $\mathcal{C}^{\otimes} \to \mathbb{F}_*$ is <u>cartesian</u> if

- the unit $\mathbb{1}$ is terminal in $\mathcal{C}$
- For x, y, the maps

$$\begin{array}{ccccc} \mathbb{1} \otimes y & \leftarrow & x \otimes y & \rightarrow & x \otimes \mathbb{1} \\ \text{IS} & & & & \text{IS} \\ y & & & & x \end{array}$$
exhibit $x \otimes y$ as $x \times y$.

Th (Lurie) If $\mathcal{C}$ is an ∞-cat w/ finite products, then $\exists!$ cart. sym. mon. str on $\mathcal{C}$ & $\mathrm{Alg}_{\mathcal{O}}(\mathcal{C}^{\times}) \simeq \mathrm{Mon}_{\mathcal{O}}(\mathcal{C})$.

Variant (cocomplete tensor product)

$\widehat{\mathrm{Cat}}_\infty$ = large ∞-cats

$\widehat{\mathrm{Cat}}_\infty^{cc}$ = cocomplete + functors preserves small colimits

$$\widehat{\mathrm{Cat}}_\infty^{cc,\otimes} \subset \widehat{\mathrm{Cat}}_\infty^{\times}$$

$$\widehat{\mathrm{Cat}}_\infty^{\times} \longrightarrow \mathbb{F}_*$$

Obj lists of cocomplete ∞-cats

mors given by list of functors

$\mathcal{C}_1 \times \mathcal{C}_2 \times \cdots \times \mathcal{C}_n \xrightarrow{F} \mathcal{D}$ s.t. F preserves colimits in each variable.

Easy to check $\widehat{\mathrm{Cat}}_\infty^{cc,\otimes}$ is an ∞-operad.

Want. It is sym mon. w/ colim-pres. $\mathcal{C} \otimes \mathcal{D} \longrightarrow \mathcal{E}$

$= \mathcal{C} \times \mathcal{D} \longrightarrow \mathcal{E}$ preserving colimits in each var.

Lecture 4. Envelopes of ∞-operads

SMCat_∞ = sym. mon. ∞-cats + strong sym. mon. functors

$\mathrm{SMCat}_\infty \longrightarrow \mathrm{Opd}_\infty$ has a left adjoint Env = sym. mon. envelope

Prop $(\mathcal{C}, \mathcal{C}_L, \mathcal{C}_R)$ ∞-cat w/ fact. syst. then $\mathrm{Ar}_R(\mathcal{C}) \xrightarrow{ev_1} \mathcal{C}$ is a cocart. fib w/ cocart mors

$$\begin{tikzcd} \cdot \arrow[r, "L"] \arrow[d, "R"'] & \cdot \arrow[d, "R"] \\ \cdot \arrow[r] & \cdot \end{tikzcd}$$

$\mathrm{Ar}_R(\mathcal{C}) \subset \mathrm{Ar}(\mathcal{C})$ full subcat on arrows in $\mathcal{C}_R$

- get $\mathcal{C} \longrightarrow \mathrm{Cat}_\infty$
 $x \longmapsto (\mathcal{C}_R)_{/x}$

- Can show for $\mathcal{E} \xrightarrow{p} \mathcal{C}$ s.t. $\mathcal{E}$ has p-cocart lifts of $\mathcal{C}_L$, then

$\mathcal{E} \underset{\mathcal{C},\, ev_0}{\times} Ar_R(\mathcal{C}) \xrightarrow[\text{via } ev_1]{} \mathcal{C}$ is a cocart fib. w/ cocart mors of the form

$$\begin{array}{ccc} e & \xrightarrow{\bar{f}} & e' \text{ cocart. over } f \\ pe & \xrightarrow[L]{f} & z \\ R\downarrow & & \downarrow R \\ x & \longrightarrow & y \end{array}$$

This is free in the sense that for $\mathcal{F} \xrightarrow{q} \mathcal{C}$ cocart fib, then

given
$$\begin{tikzcd} \mathcal{E} \arrow[rr, "F"] \arrow[dr, "p"'] & & \mathcal{F} \arrow[dl, "q"] \\ & \mathcal{C} & \end{tikzcd}$$
s.t. F preserves cocart lifts of $\mathcal{C}_L$

$\exists!$ ext'n to
$$\begin{tikzcd} \mathcal{E} \underset{\mathcal{C}}{\times} Ar_R(\mathcal{C}) \arrow[rr, "F'"] \arrow[dr] & & \mathcal{F} \arrow[dl] \\ & \mathcal{C} & \end{tikzcd}$$
s.t. F' preserves all cocart mors.

$$\left(x,\ px \xrightarrow[R]{g} c\right) \longmapsto g_! F(x) \in \mathcal{F}$$

so for $\mathcal{O} \to \mathbb{F}_*$ ∞-operad has cocart lifts of inerts, - so $Env(\mathcal{O}) := \mathcal{O} \underset{\mathbb{F}_*}{\times} Ar_{act}(\mathbb{F}_*)$

has required property. provided $Env(\mathcal{O})$ is sym. mon.

$$Env(\mathcal{O})_{\langle n \rangle} = \mathcal{O} \underset{\mathbb{F}_*}{\times} \left(\mathbb{F}_*^{act}\right)_{/\langle n \rangle} \longrightarrow \prod_{i=1}^{n} \mathcal{O}^{act}$$

$Env(\mathcal{O})_{\langle 1 \rangle} = \mathcal{O}^{act}$ Subcat of maps over actives in $\mathbb{F}_*$

Ob of $Env(\mathcal{O})_{\langle n \rangle}$ is $X \in \mathcal{O}$, $\underset{\langle m \rangle}{\overset{\|}{p(X)}} \xrightarrow[act]{\varphi} \langle n \rangle$

Using
$$\begin{tikzcd} \langle m \rangle \arrow[r, "\varphi"] \arrow[d, "int"'] & \langle n \rangle \arrow[d, "\rho_i"] \\ \langle m_i \rangle \arrow[r, "act"'] & \langle 1 \rangle \end{tikzcd} \rightsquigarrow \begin{array}{c} X \\ \downarrow \\ X_i \text{ over } \langle m_i \rangle \end{array}$$
gives equiv. to be sym. mon

Passed

$\mathrm{Env}(\mathcal{O}) \longrightarrow \mathbb{F}_*$ is a sym. mon. str. on $\mathcal{O}^{act}$

- objs $(x_1, \dots x_n)$ $x_i \in \mathcal{O}_{\langle 1\rangle}$
- mors lists of multimors from $\mathcal{O}$
- Sym. mon. str. = concatenating lists.

Examples $\mathrm{Env}(\mathbb{F}_*) = (\mathbb{F}, \amalg)$

Have adjunction
$$\begin{tikzcd} \mathrm{SMCat}_\infty \arrow[r] & \mathrm{Opd}_\infty \arrow[l, bend right, "\mathrm{Env}"'] \arrow[l, phantom, "\perp"] \end{tikzcd}$$

$$\leadsto \begin{tikzcd} \mathrm{SMCat}_\infty/\mathbb{F}^{\amalg} \arrow[r, "\mathcal{U}"'] & \mathrm{Opd}_\infty \arrow[l, bend right, "\mathrm{Env}"'] \arrow[l, phantom, "\perp"] \end{tikzcd}$$ w/ $\mathcal{U}$ given by

$$\begin{tikzcd} \mathcal{U}(\mathcal{C}^\otimes/\mathbb{F}^{\amalg}) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{C}^\otimes \arrow[d] \\ \mathbb{F}_* \arrow[r] & \mathbb{F}^{\amalg} \end{tikzcd}$$

$\mathcal{U}\,\mathrm{Env} \simeq \mathrm{id}$ since

$$\begin{tikzcd} \mathcal{U}\mathrm{Env}(\mathcal{O}) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathrm{Env}(\mathcal{O}) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{O} \arrow[d] \\ \mathbb{F}_* \arrow[r] \arrow[rr, bend right, "\mathrm{id}"'] & \mathrm{Ar}_{act}(\mathbb{F}_*) \arrow[r] & \mathbb{F}_* \end{tikzcd}$$

$\Rightarrow$ Env is fully faithful as functor to $\mathrm{SMCat}_\infty/\mathbb{F}^{\amalg}$

Prop (Barkan - Steinebrunner) Env identifies Opd_∞ w/ full subcat of $\mathrm{SMCat}_\infty/\mathbb{F}^{\amalg}$ on $\mathcal{C}^\otimes \longrightarrow \mathbb{F}^{\amalg}$ that are equifibred:

for every $\langle n\rangle \xrightarrow{\varphi} \langle m\rangle$ active,
$$\begin{tikzcd} \mathcal{C}^\otimes_{\langle n\rangle} \arrow[r, "\varphi_!"] \arrow[d] & \mathcal{C}^\otimes_{\langle m\rangle} \arrow[d] \\ \mathbb{F}_{/\underline{n}} \arrow[r, "\varphi_!"] & \mathbb{F}_{/\underline{m}} \end{tikzcd}$$
is a pull back.

$$\Leftrightarrow \begin{tikzcd} \mathcal{C} \times \mathcal{C} \arrow[r, "\otimes"] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{C} \arrow[d] \\ \mathbb{F} \times \mathbb{F} \arrow[r, "\amalg"] & \mathbb{F} \end{tikzcd}$$

[i.e. $X \in \mathcal{C}$ over $\underline{n} \amalg \underline{m}$, then $X = X_1 \otimes X_2$ w/ X_1 over $\underline{n}$, X_2 over $\underline{m}$ in a unique way]

$U(\mathcal{C} \to \mathbb{F})$ suboperad of $\mathcal{C}^{\otimes}$ on objs that lie over $\underline{1} \in \mathbb{F}$.

- colours $\mathcal{C}_{\langle 1 \rangle}$
- multimor $(X_1, \dots, X_n) \to Y$ is mor $X_1 \otimes \cdots \otimes X_n \to Y$ over $\underline{1} \sqcup \cdots \sqcup \underline{1} \to \underline{1}$

in $\mathcal{C}$.

<u>Variant</u> (H-Kock) A PROP is a sym. mon. ∞-cat s.t. underlying sym mon. ∞-gpd is <u>free</u>.

$$\begin{tikzcd}
\mathrm{PROP}_\infty \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathrm{SMCat}_\infty \arrow[d] \\
\mathcal{S} \arrow[r, "\mathrm{Free}"] & \mathrm{CAlg}(\mathcal{S})
\end{tikzcd}$$

Envelopes are PROPs, so get $\mathrm{Env} : \mathrm{Opd}_\infty \to \mathrm{PROP}_\infty$.

This is also fully faithful & can describe image

Barkan - Steinebrunner show $\mathrm{Env} : \mathrm{Opd}_\infty \to \mathrm{SMCat}_\infty$ is a subcat inc.

<u>Day convolution</u>

$\mathcal{C}$ small sym. mon. cat. $\mathcal{D}$ cocomplete sym. mon. cat s.t. $\otimes$ preserves colimits in each variable. Then $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$ has a sym. mon str. where $F \otimes G$ is

$$\begin{tikzcd}
\mathcal{C} \times \mathcal{C} \arrow[r, "F \times G"] \arrow[d, "\otimes_{\mathcal{C}}"'] & \mathcal{D} \times \mathcal{D} \arrow[d, "\otimes_{\mathcal{D}}"] \\
\mathcal{C} \arrow[r, dashed, "\exists \mathrm{LKE} = F \otimes G"] & \mathcal{D}
\end{tikzcd}$$

i.e. $(F \otimes G)(c) = \operatorname{colim}_{(x, y,\, x \otimes y \to c)} F(x) \otimes G(y)$

Comm. algs in $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$ are lax sym. mon functors $\mathcal{C} \to \mathcal{D}$.

$F \otimes F \to F$ amounts to giving for $c \in \mathcal{C}$ and $x \otimes y \to c$, compatible maps

$F(x) \otimes F(y) \to F(c)$

- generated by $F(x) \otimes F(y) \to F(x \otimes y)$

Th (Glasman, Lurie) $\mathcal{C}$ small sym. mon. ∞-cat, $\mathcal{O}$ an ∞-opd. Then $\exists$ an ∞-opd $\mathrm{Day}_{\mathcal{C}}(\mathcal{O})$ s.t. $\mathrm{Alg}_{\mathcal{P}}(\mathrm{Day}_{\mathcal{C}}(\mathcal{O})) \simeq \mathrm{Alg}_{\mathcal{P} \times_{\mathbb{F}_*} \mathcal{C}^{\otimes}}(\mathcal{O})$.

If $\mathcal{O} = \mathcal{D}^{\otimes}$ is sym. mon w/ $\mathcal{D}$ cocomplete & $\otimes$ preserves colimits in each variable, then $\mathrm{Day}_{\mathcal{C}}(\mathcal{D}^{\otimes})$ is sym. mon.

- $\mathrm{Day}_{\mathcal{C}}(\mathcal{O})_{\langle 1 \rangle} \simeq \mathrm{Fun}(\mathcal{C}, \mathcal{O}_{\langle 1 \rangle})$

- $\mathrm{Day}_{\mathcal{C}}(\mathcal{O})$ is internal Hom for cartesian product in Opd_{∞}

  - __note__ Opd_{∞} is __not__ cartesian closed

- Can extend to $\mathcal{C}^{\otimes}$ is "symmetric promonoidal" (Barwick - Glasman - Shah, Shah, Hinich)

__Variant__ Can show $\mathrm{PSh}: \mathrm{Cat}_{\infty} \longrightarrow \widehat{\mathrm{Cat}}_{\infty}^{cc}$ is ~~lax~~ sym. mon

w.r.t. $\times$ $\otimes$

For $\mathcal{C}$ sym mon, this means $\mathrm{PSh}(\mathcal{C})$ is sym mon w/ $\otimes$ preserving colimit in each var.

Glasman shows this agrees w/ Day convolution from $\mathcal{C}$ & $(\mathcal{S}, \times)$

This leads to another univ property

$$\left\{ \begin{matrix} \text{colim. preserving} \\ \text{sym mon } \mathrm{PSh}(\mathcal{C}) \to \mathcal{D} \end{matrix} \right\} \simeq \left\{ \begin{matrix} \text{sym mon} \\ \mathcal{C} \to \mathcal{D} \end{matrix} \right\}$$

$\mathcal{C} \rightsquigarrow \mathrm{Day}_{\mathcal{C}^{op}}(\mathcal{S}^{\times})$

$\mathcal{C} \rightsquigarrow \mathrm{PSh}(\mathcal{C})$ c-alg in $\widehat{\mathrm{Cat}}_{\infty}^{cc}$

__Examples__ · $[1] = 0 \to 1$ has sym mon str. $i \otimes j = \min(i, j)$

If $\mathcal{C}$ sym mon ∞-cat compatible w/ ~~coproducts~~ pushouts, get Day convolution on $\mathrm{Ar}(\mathcal{C})$

w/ $(x \to y) \otimes (x' \to y') = x \otimes y' \underset{x \otimes x'}{\amalg} y \otimes x' \to y \otimes y'$.

For $\mathcal{C}^\times$, this localises to sym. mon. on $\mathcal{C}_* = \mathcal{C}_{*/} \subset Ar(\mathcal{C})$ w/ left adjoint

$L : Ar(\mathcal{C}) \to \mathcal{C}_*$, $(x \to y) \mapsto (* \to y/x = y \underset{x}{\amalg} *)$

[$\mathcal{C}_0 \underset{\hookrightarrow}{\overset{L}{\leftarrow}} \mathcal{C}$ $\rightsquigarrow$ smash product on $\mathcal{C}_*$.

$L(x \to y)$ equiv, $L(c \otimes x \to c \otimes y)$ equiv

$\to$ $\mathcal{C}_0$ is sym mon w/ tensor $L(- \otimes -)$]

- $CMon(\mathcal{C}) \subset Fun(\mathbb{F}_*, \mathcal{C})$ – has Day conv from $\wedge$ on $\mathbb{F}_*$ & $\times$ on $\mathcal{C}$.

This localises to give $\otimes$ on $CMon(\mathcal{C})$ & $CGrp(\mathcal{C})$ [Gepner – Groth – Nikolaus, Ben Moshe – Schlank]

- Spectra: can view a spectrum as a functor $\mathcal{S}_*^{fin} \xrightarrow{F} \mathcal{S}$ that is

<u>reduced</u> $F(*) \simeq *$ & <u>excisive</u> $F(\text{pushout})$ is a pullback

$F(S^n) \simeq \Omega F(S^{n-1})$

- $Fun(\mathcal{S}_*^{fin}, \mathcal{S})$ has Day convolution from $\wedge$ on $\mathcal{S}_*^{fin}$, $\times$ on $\mathcal{S}$

localises to Sp to give smash product.

<u>Lecture 5</u>. <u>Boardman – Vogt tensor products</u>

Consider operad $\mathcal{O}$ w/ one obj. $\mathcal{C}$ sym mon cat

Then $Alg_{\mathcal{O}}(\mathcal{C})$ is sym. mon via $\otimes$ from $\mathcal{C}$

$A, B \in Alg_{\mathcal{O}}(\mathcal{C})$ can make $A \otimes B$ $\mathcal{O}$-alg via

$$\mathcal{O}(n) \underset{\Sigma_n}{\times} (A \otimes B)^{\otimes n} \underset{\substack{\uparrow \\ \text{uses diagonal!}}}{\longrightarrow} \left(\mathcal{O}(n) \underset{\Sigma_n}{\times} A^{\otimes n}\right) \otimes \left(\mathcal{O}(n) \underset{\Sigma_n}{\times} B^{\otimes n}\right) \to A \otimes B$$

Pyhhes.

More generally, for $\mathcal{P}$ coloured operad, $\mathrm{Alg}_{\mathcal{O}}(\mathcal{P})$ has an operad str.

alg. multimor $(A_1, \dots, A_n) \to B$ = multimor. in $\mathcal{P}$ s.t. for $\alpha \in \mathcal{O}(k)$ the square

$$\begin{tikzcd}
(\underbrace{A_1, \dots, A_1}_{k}, \dots, \underbrace{A_n, \dots, A_n}_{k}) \arrow[r, "\alpha_{A_1}, \dots, \alpha_{A_n}"] \arrow[d, "\wr"] & (A_1, \dots, A_n) \arrow[d, "f"] \\
(A_1, \dots, A_n, \dots, A_1, \dots, A_n) \arrow[d, "(\underbrace{f, \dots, f}_{k})"] & B \\
(B, \dots, B) \arrow[ur, "\alpha_B"] &
\end{tikzcd}$$

commutes

This operad str. on algebras in adjoint the Boardman-Vogt tensor product.

$$\mathcal{O} \longrightarrow \mathrm{Alg}_{\mathcal{P}}(\mathcal{Q}) \quad \longleftrightarrow \quad \mathcal{O} \otimes \mathcal{P} \longrightarrow \mathcal{Q}$$

Can describe functors out of $\mathcal{O} \otimes \mathcal{P}$ as <u>bifunctors</u> of operad

<u>Def</u>. A <u>bifunctor</u> of ∞-operad $(\mathcal{O}, \mathcal{P}) \to \mathcal{Q}$ is a comm. square

$$\begin{tikzcd}
\mathcal{O} \times \mathcal{P} \arrow[r, "F"] \arrow[d] & \mathcal{Q} \arrow[d] \\
\mathbb{F}_* \times \mathbb{F}_* \arrow[r, "\wedge"] & \mathbb{F}_*
\end{tikzcd}$$

s.t. F takes pairs of inert cocart mors to inert cocart. in $\mathcal{Q}$.

- Can show bifunctors are rep'd by an ∞ opd $\mathcal{O} \otimes \mathcal{P}$
  - the Boardman-Vogt tensor.
- Can construct ∞-operad str. on $\mathrm{Alg}_{\mathcal{O}}(\mathcal{P})$ s.t. $\mathcal{Q} \longrightarrow \mathrm{Alg}_{\mathcal{O}}(\mathcal{P})$

  $\simeq$ bifunctors $\mathcal{O} \times \mathcal{Q} \longrightarrow \mathcal{P}$.

<u>Th</u>. (Lurie, Barwick-Steinebrunner) The BV $\otimes$ gives a sym. mon. str. on Opd_∞.

($\wedge$ on $\mathbb{F}_*$ $\rightsquigarrow$ $\otimes$ in SMCat_∞ $\rightsquigarrow$ induces BV on Opd_∞)

Def The little disc operad E_n is a 1-ob. operad in Top w/

$E_n(k)$ = top. sp. of disjoint embeddings of k n-discs in a bigger one

$\rightsquigarrow$ ∞-opd $\mathbb{E}_n$

$E_n(k) \simeq$ Config. space of k-pts in $\mathbb{R}^n$

- $\mathbb{E}_1 \simeq$ Assoc
- $\mathbb{E}_\infty = \operatorname{colim}_{n\to\infty} \mathbb{E}_n$ is $\mathbb{E}_\infty$ = Comm

$\mathbb{E}_n \longrightarrow \mathbb{E}_{n+1}$

cross w/ $[0,1]$

Th (Dunn additivity, Lurie) $\mathbb{E}_n \otimes \mathbb{E}_m \simeq \mathbb{E}_{n+m}$

or $\mathbb{E}_n \simeq \mathbb{E}_1^{\otimes n}$

I.e. an $\mathbb{E}_n$-alg is a str. w/ n compatible associative multiplications.

Eckmann-Hilton : In a 1-cat, $\mathbb{E}_2$-algs & higher = comm. algs.

Def An ∞-cat $\mathcal{C}$ is an $\underline{(n,1)\text{-cat}}$ if $\mathcal{C}(x,y)$ is an $\underbrace{(n-1)\text{-type}}_{\pi_i \text{ vanish for } i>n-1}$ for all $x, y \in \mathcal{C}$

Th (Lurie) If $\mathcal{C}$ is a sym. mon. $(n,1)$-cat, then

$\mathrm{Alg}_{\mathbb{E}_k}(\mathcal{C}) \simeq \mathrm{CAlg}(\mathcal{C})$ if $k \geq n+1$.

Cor. (Baez-Dolan Stabilization)

k-uply monoidal n-cats

$= \mathbb{E}_1^{\otimes k}$-algs in Cat_n [∞-cat of n-cats]

are equiv. to sym mon n-cats if $k \geq n+2$.

Proof. Cat_n is $(n+1,1)$-cat.

Free algebras & operadic Kan ext'ns

$\mathcal{O}$ a 1-obj operad, $\mathcal{C}$ sym mon cat (compatible w/ colimits over gpds)

The free $\mathcal{O}$-alg on $X \in \mathcal{C}$ is

$$\underbrace{\coprod_{n=0}^{\infty} \mathcal{O}(n) \underset{\Sigma_n}{\times} X^{\otimes n}}_{\coprod_{\mathcal{O}(n)} X^{\otimes n} / \text{diagonal } \Sigma_n\text{-action}}$$

For $f: \mathcal{O} \longrightarrow \mathcal{P}$ have $f^*: \mathrm{Alg}_{\mathcal{P}}(\mathcal{C}) \longrightarrow \mathrm{Alg}_{\mathcal{O}}(\mathcal{C})$

If $\mathcal{C}$ is compatible w/ certain colimits, then f^* has a left adjoint $f_!$ given by an explicit formula

<u>Notation</u> For
$$\begin{tikzcd} \mathcal{O} \arrow[r, "f"] \arrow[dr] & \mathcal{P} \arrow[d] \\ & \mathbb{F}_* \end{tikzcd}$$
mor of ∞-operads, $p \in \mathcal{P}$,

we write $\mathcal{O}^{act}_{/p} = \mathcal{O}^{act} \underset{\mathcal{P}^{act}}{\times} \mathcal{P}^{act}_{/p}$ obj: $x \in \mathcal{O}$, $f(x) \underset{\text{active}}{\longrightarrow} p$

<u>Def</u> A sym mon ∞-cat $\mathcal{C}^{\otimes}$ is <u>compatible w/ K-indexed colimits</u>

if $\mathcal{C}$ has K-colimits & $\otimes$ preserves them in each variable.

<u>Th</u>. (Lurie) $f: \mathcal{O} \to \mathcal{P}$ mor of ∞-gpds, $\mathcal{C}^{\otimes}$ sym mon ∞-cat, compatible w/ $\mathcal{O}^{act}_{/p}$-colimits for all $p \in \mathcal{P}$. Then $f^*: \mathrm{Alg}_{\mathcal{P}}(\mathcal{C}) \to \mathrm{Alg}_{\mathcal{O}}(\mathcal{C})$ has a left adjoint $f_!$ s.t. for $p \in \mathcal{P}_{\langle 1 \rangle}$

$$(f_! A)(p) \simeq \underset{\substack{(x,\, fx \xrightarrow{\alpha} p) \in \mathcal{O}^{act}_{/p} \\ \langle n \rangle \xrightarrow{\pi(\alpha)} \langle 1 \rangle}}{\mathrm{colim}} \underbrace{\pi(\alpha)_!\, A(\alpha)}$$

i.e. for $X = (x_1, \dots x_n)$ w/ $x_i \in \mathcal{O}_{\langle 1 \rangle}$, $\pi: \mathcal{P} \longrightarrow \mathbb{F}_*$ this is $A(x_1) \otimes \dots \otimes A(x_n)$

<u>Observation</u> $\mathcal{O}$ ∞-opd, then so is $\mathcal{O}^{int}$ (subcat of inert cocart mors)

Then $\mathcal{O}^{\mathrm{int}}_{\langle 1\rangle} \simeq \mathcal{O}^{\simeq}_{\langle 1\rangle}$, $\mathrm{Alg}_{\mathcal{O}^{\mathrm{int}}}(\mathcal{C}) \simeq \mathrm{Fun}(\mathcal{O}^{\simeq}_{\langle 1\rangle}, \mathcal{C})$

For $\mathcal{O}^{\mathrm{int}} \to \mathcal{O}$, $(\mathcal{O}^{\mathrm{int}})^{\mathrm{act}}_{/x} \simeq (\mathcal{O}^{\mathrm{act}}_{/x})^{\simeq} =: \mathrm{Act}_{\mathcal{O}}(x)$

<u>Cor.</u> $V_{\mathcal{O}} : \mathrm{Alg}_{\mathcal{O}}(\mathcal{C}) \to \mathrm{Alg}_{\mathcal{O}^{\mathrm{int}}}(\mathcal{C}) \simeq \mathrm{Fun}(\mathcal{O}^{\simeq}_{\langle 1\rangle}, \mathcal{C})$ has a left adjoint

$\mathrm{Free}_{\mathcal{O}}$.

For X, $\mathcal{O}^{\simeq}_{\langle 1\rangle} \to \underset{(y_1,\dots y_n)\to y}{\mathrm{colim}}\; X(y_1)\otimes\cdots\otimes X(y_n)$

over $\mathrm{Act}_{\mathcal{O}}(y)$

<u>Observation</u> Suppose $\mathcal{O}^{\simeq}_{\langle 1\rangle} = *$. Then

$$\mathrm{Act}_{\mathcal{O}}(*) \to \mathrm{Act}_{\mathbb{E}_\infty}(\langle 1\rangle) \simeq (\mathbb{F}^{\mathrm{act}}_{*/\langle 1\rangle})^{\simeq} \simeq \mathbb{F}^{\simeq} = \coprod_n B\Sigma_n$$

Recover $\mathcal{O}(n) \curvearrowleft \Sigma_n$ as fibre over pt in $B\Sigma_n$

i.e. $\mathrm{Act}_{\mathcal{O}}(*) = \coprod_n \mathcal{O}(n)_{h\Sigma_n}$

$\to$ can compute colim over $\mathrm{Act}_{\mathcal{O}}(*)$ as iterated colim

$$\underset{\mathrm{Act}_{\mathcal{O}}(*)}{\mathrm{colim}}\, F \simeq \underset{\coprod B\Sigma_n}{\mathrm{colim}}\; \underset{\mathcal{O}(n)}{\mathrm{colim}}\, F \simeq \coprod \underset{B\Sigma_n}{\mathrm{colim}}\; \underset{\mathcal{O}(n)}{\mathrm{colim}}\, F$$

$$\Rightarrow \mathrm{Free}_{\mathcal{O}} X(*) \simeq \coprod_n \underbrace{\underset{B\Sigma_n}{\mathrm{colim}}\; \underset{\mathcal{O}(n)}{\mathrm{colim}}\; X(*)^{\otimes n}}_{(\mathcal{O}(n)\times X(*)^{\otimes n})_{h\Sigma_n}}$$

More generally, if $\mathcal{O}^{\simeq}_{\langle 1\rangle}$ is conn'd, classical formula is left adjoint to

$\mathrm{Alg}_{\mathcal{O}}(\mathcal{C}) \to \mathrm{Fun}(\mathcal{O}^{\simeq}_{\langle 1\rangle}, \mathcal{C}) \to \mathcal{C}$

Exer If $\mathcal{C}^\times$ is cartesian monoidal, then $f_! : \mathrm{Alg}_{\mathcal{O}}(\mathcal{C}^\times) \to \mathrm{Alg}_{\mathcal{P}}(\mathcal{C}^\times)$ is computed by restricting ordinary LKE $\mathrm{Fun}(\mathcal{O}, \mathcal{C}) \xrightarrow[f_!]{} \mathrm{Fun}(\mathcal{P}, \mathcal{C})$ to monoids.

General proofs.

- Lurie
- Chu-H. use Day convolution to reduce to monoids
- Nardin-Shah

$\mathcal{C}^\otimes$ sym mon compatible w/ small colimits. $\mathcal{O}$ small ∞-opd.

$\mathrm{Free}_{\mathcal{O}} : \mathrm{Fun}(\mathcal{O}_{\langle 1 \rangle}^{\simeq}, \mathcal{C}) \rightleftarrows \mathrm{Alg}_{\mathcal{O}}(\mathcal{C}) : U_{\mathcal{O}}$ (with $\perp$)

Then

- $U_{\mathcal{O}}$ is conservative
- $\mathrm{Alg}_{\mathcal{O}}(\mathcal{C})$ has K-indexed limits if $\mathcal{C}$ does
- $\mathrm{Alg}_{\mathcal{O}}(\mathcal{C})$ has sifted colimits & $U_{\mathcal{O}}$ preserves these
- $\mathrm{Free}_{\mathcal{O}} \dashv U_{\mathcal{O}}$ is monadic.
- $\mathrm{Alg}_{\mathcal{O}}(\mathcal{C})$ has small colimits
- If $\mathcal{C}$ presentable, so is $\mathrm{Alg}_{\mathcal{O}}(\mathcal{C})$.
- Can recover $\mathcal{O}$ from free alg monad w/ $\mathcal{C} = \mathcal{S}$.